



SRH University Heidelberg

**Reliability Monitoring and Cost-Sensitive Selective
Prediction for Credit Risk under Distribution Shift**

Master Thesis Exposé

by

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1 Introduction

Credit risk assessment plays a fundamental role in modern financial systems. Banks and financial institutions rely on predictive models to estimate the probability that a borrower will default on a loan. Accurate credit risk prediction enables financial institutions to reduce financial losses, improve risk management, and ensure the stability of lending systems.

Traditionally, statistical methods such as logistic regression and discriminant analysis have been widely used in credit scoring systems [5]. These models are valued for their interpretability and simplicity. However, the rapid growth of financial data and computational capabilities has led to the increasing adoption of machine learning techniques in credit risk modelling [6].

Machine learning algorithms such as decision trees, random forests, gradient boosting machines, and neural networks have demonstrated strong predictive performance in credit risk prediction tasks [6], [16]. These models can capture complex nonlinear relationships within financial datasets and often outperform traditional statistical methods.

Despite their predictive advantages, deploying machine learning models in real-world financial systems introduces several important challenges. One major issue is the presence of distribution shifts, where the statistical properties of new data differ from those observed during model training. Such shifts can occur due to economic fluctuations, regulatory changes, or evolving borrower behaviour [3], [10].

Distribution shifts can significantly degrade model performance and lead to unreliable predictions. In high-stakes domains such as financial decision-making, unreliable predictions may result in substantial financial losses and increased operational risks.

Another critical challenge in credit risk prediction is the asymmetric cost of prediction errors. Approving a loan for a borrower who later defaults may result in significant financial losses for financial institutions. Conversely, rejecting a reliable borrower typically results only in lost business opportunities. Most machine learning models optimize overall classification accuracy without considering the differing economic consequences of misclassification errors [7], [9], [14].

Furthermore, many predictive models always produce a prediction regardless of their confidence level. When models encounter unfamiliar or uncertain data, forcing a prediction may lead to unreliable or risky decisions [4], [12].

To address these challenges, recent research has explored several directions, including cost-sensitive learning, uncertainty estimation, and selective prediction [4], [7], [9], [12], [14], [16]. However, these approaches are often studied independently and are rarely integrated into a unified framework for credit risk modelling under distribution shift.

Therefore, this thesis proposes a reliability-aware machine learning framework that integrates **reliability monitoring, cost-sensitive learning, and selective prediction** to improve the robustness and trustworthiness of credit risk prediction systems [2], [7], [12].

1.1 Motivation

Machine learning models are increasingly being used in financial institutions to automate credit approval decisions and support risk management processes. While these models often demonstrate high predictive performance in controlled environments, their reliability in dynamic real-world settings remains a major concern.

Financial systems are strongly influenced by macroeconomic conditions. Changes in economic indicators such as unemployment rates, inflation, or market volatility may alter borrower behaviour and lead to shifts in data distributions [3], [10].

Recent developments in Reliable Artificial Intelligence emphasize the importance of building systems that remain robust and trustworthy under uncertain and changing conditions. Reliability monitoring techniques aim to detect when models become unreliable due to distribution shifts or concept drift [3], [11].

In addition, financial decision-making requires careful consideration of the economic impact of prediction errors. Cost-sensitive learning approaches incorporate the financial consequences of different types of misclassification errors into the model training process [7], [9], [14].

Another promising approach is selective prediction, also known as prediction with abstention. Selective prediction allows models to refrain from making predictions when the confidence level is low. Instead of forcing uncertain predictions, the model can defer these cases to human experts or additional evaluation processes [4], [12].

Integrating these approaches can significantly improve the reliability and practical applicability of machine learning systems in financial decision-making environments [2], [7], [12].

1.2 Problem Statement

Most existing credit risk prediction systems assume that the distribution of training data remains stable over time. However, this assumption is often violated in real-world financial environments [3], [10].

Distribution shifts may occur due to economic crises, regulatory changes, evolving borrower demographics, or changes in market conditions. When such shifts occur, machine learning models trained on historical data may experience performance degradation, poor calibration, and increased prediction uncertainty [3], [10].

At the same time, traditional machine learning models treat all classification errors equally. In credit risk prediction, however, the financial consequences of different errors vary significantly. Approving a loan for a borrower who later defaults may cause large financial losses, while rejecting a reliable borrower leads only to missed opportunities [7], [14].

Another limitation of many predictive models is that they always produce a prediction regardless of their level of confidence. When models encounter unfamiliar or uncertain data, forcing a prediction may lead to unreliable decisions [4], [12].

Although previous studies have explored cost-sensitive learning and uncertainty estimation, limited research has investigated the integration of **distribution shift analysis, reliability monitoring, cost-sensitive learning, and selective prediction** within a unified framework for credit risk modelling [2], [7], [12].

This research aims to address this gap by developing and evaluating a reliability-aware credit risk prediction framework.

1.3 Research Questions

This study aims to answer the following research questions:

RQ1

How does distribution shift affect the reliability and calibration of machine learning models used for credit risk prediction?

RQ2

Can cost-sensitive learning methods reduce financial losses caused by misclassification errors?

RQ3

Can selective prediction improve decision reliability by allowing models to abstain from uncertain predictions?

RQ4

How effectively can reliability monitoring techniques detect model degradation under distribution shifts?

RQ5

What combination of cost-sensitive learning and selective prediction provides the best balance between predictive performance and financial risk?

2 State of the Art

2.1 Credit Risk Modelling

Credit risk modelling has long been a fundamental topic in financial risk management. Early credit scoring systems relied primarily on **statistical techniques** such as **logistic regression** and **discriminant analysis** [5]. These models remain widely used due to their **interpretability, simplicity, and ease of implementation**. They allow financial institutions to understand how each feature (e.g., income, debt ratio, or credit history) contributes to the predicted risk of default.

However, traditional statistical models have limitations. They often assume linear relationships between variables, struggle to capture complex patterns, and are sensitive to outliers. In real-world financial datasets, relationships between borrower attributes and default risk are often **nonlinear and dynamic**, motivating the use of more sophisticated machine learning approaches.

2.2 Machine Learning for Credit Risk Prediction

Recent advances in machine learning have enabled the development of models capable of capturing **nonlinear interactions and complex dependencies** in borrower data. Techniques such as **Random Forests, Gradient Boosting Machines (e.g., XGBoost, LightGBM), and Neural Networks** have been applied extensively in credit risk prediction [6], [16].

Random Forests are ensemble methods that combine multiple decision trees to improve predictive accuracy and reduce overfitting. Gradient Boosting Machines iteratively combine weak learners to optimize prediction performance, making them highly effective for structured financial data. Neural Networks, particularly deep learning models, can learn hierarchical feature representations, capturing subtle patterns that traditional models might miss.

Despite their advantages, machine learning models face challenges in real-world deployment. They can be **sensitive to changes in data distributions**, lack interpretability, and often do not account for the **asymmetric costs of misclassification** in financial decisions.

2.3 Cost-Sensitive Learning

In credit risk prediction, the financial consequences of prediction errors are not uniform. **Approving a loan for a borrower who defaults** can result in substantial financial losses,

whereas **rejecting a reliable borrower** mainly results in a missed opportunity. Standard machine learning models treat all errors equally, which is inadequate in financial settings.

Cost-sensitive learning addresses this issue by **incorporating misclassification costs into the training process** [7], [9], [14]. This can be achieved using **weighted loss functions** or **cost matrices**, where the model penalizes false positives (high-risk approvals) more than false negatives (safe rejections). By explicitly integrating the economic impact of errors, cost-sensitive models aim to **minimize expected financial losses rather than overall misclassification rate**.

For example, a logistic regression model can be modified with a weighted loss function to reflect the higher cost of risky approvals. Similarly, tree-based models can adjust their splitting criteria based on cost considerations.

2.4 Distribution Shift and Concept Drift

A major challenge in deploying credit risk models is that **data distributions change over time**, a phenomenon known as **distribution shift** or **concept drift** [3], [10]. Changes in economic conditions, regulatory policies, market behavior, or borrower demographics can cause the statistical properties of new data to deviate from historical data used for training.

Such shifts can **degrade model performance**, reduce calibration, and increase uncertainty in predictions. For example, a model trained during a period of low unemployment may underestimate default risk during a recession.

To address this, it is crucial to **detect distribution shifts** and **adapt models accordingly**. Techniques such as **Population Stability Index (PSI)**, **drift detection algorithms**, and **periodic recalibration** can be used to monitor and maintain reliability over time.

2.5 Selective Prediction

Selective prediction, or **prediction with abstention**, allows models to **avoid making predictions when the confidence level is low** [4], [12]. Instead of forcing a decision, the model can defer uncertain cases to human experts or additional evaluation processes.

This approach is particularly useful in financial decision-making, where an incorrect prediction can have high financial costs. Selective prediction improves overall **decision reliability**, ensures that only high-confidence predictions are acted upon, and helps institutions **mitigate risks in uncertain or changing environments**.

Implementation strategies include using **softmax probabilities in neural networks** or **margin-based thresholds in ensemble models**. The confidence threshold can be tuned based on risk appetite or cost considerations.

3 Methodology

3.1 Data Collection

Publicly available credit risk datasets will be used:

- **Home Credit Default Risk Dataset:** Includes borrower demographics, credit history, and loan attributes.
- **German Credit Dataset:** Contains structured financial and personal attributes with labeled default outcomes.
- **Lending Club Loan Dataset:** Large-scale dataset with borrower profiles, loan characteristics, and repayment outcomes.

These datasets provide **diverse and representative scenarios** for modeling credit risk, including imbalanced classes and varying borrower behavior.

3.2 Data Preprocessing

Preprocessing is essential to ensure model performance and reliability:

- **Handling missing values:** Imputation using median, mode, or K-nearest neighbors methods.
- **Feature encoding:** Categorical variables are transformed using one-hot or target encoding.
- **Normalization/Standardization:** Ensures features are on a comparable scale.
- **Class imbalance handling:** Techniques such as **SMOTE (Synthetic Minority Oversampling Technique)** are used to address the underrepresentation of default cases.

Preprocessing ensures that models are trained on clean, representative, and balanced datasets.

3.3 Baseline Models

The following machine learning models will serve as baselines:

- **Logistic Regression:** Interpretable linear model for initial benchmarking.
- **Random Forest:** Ensemble tree-based model capturing nonlinear relationships.
- **Gradient Boosting (XGBoost/LightGBM):** High-performance ensemble models for structured data.

These baseline models provide a **reference point** for evaluating the effectiveness of cost-sensitive and selective prediction techniques.

3.4 Cost-Sensitive Learning

Cost-sensitive learning will be implemented using:

- **Weighted loss functions:** Assign higher penalties to costly misclassifications.
- **Cost matrices:** Explicitly define the financial consequences of false positives and false negatives.

This ensures that the model optimizes not only predictive accuracy but also **financial risk reduction**.

3.5 Selective Prediction

Selective prediction mechanisms will be incorporated to **allow models to abstain from low-confidence predictions**.

- Thresholds will be tuned based on risk appetite and validation performance.
- Abstained cases will be deferred for **human review** or **additional analysis**, reducing risky decisions.

3.6 Distribution Shift Simulation

To evaluate robustness, **distribution shifts will be simulated** by:

- Modifying feature distributions to mimic changing borrower behavior.

- Adjusting default rates to represent economic fluctuations or crises.

This allows testing model reliability in **dynamic and realistic scenarios**.

3.7 Reliability Monitoring

Model reliability will be assessed using multiple metrics:

- **Brier Score:** Measures calibration of probabilistic predictions.
- **Expected Calibration Error (ECE):** Quantifies the difference between predicted probabilities and actual outcomes.
- **Population Stability Index (PSI):** Detects shifts in feature distributions over time.

These metrics ensure that models are **trustworthy, well-calibrated, and robust to changes**.

3.8 Adaptive Reliability-aware Decision Framework

To further enhance decision reliability, this research proposes an adaptive reliability-aware decision mechanism that determines whether a model prediction should be accepted automatically or deferred to human experts.

Traditional predictive models always produce a prediction regardless of their confidence level or the reliability of the input data. However, in financial decision-making environments, predictions made under high uncertainty or data distribution shifts may lead to significant financial risks.

To address this issue, the proposed framework introduces a **Review Score** that combines multiple reliability signals to evaluate the trustworthiness of each prediction.

The Review Score integrates three key components:

Prediction Uncertainty: measures how confident the model is about the predicted probability of default.

Distribution Shift (Drift Score): captures how much the current input data deviates from the training data distribution.

Expected Financial Cost: estimates the potential financial loss associated with an incorrect prediction.

The combined decision score is defined as:

$$\text{ReviewScore} = \alpha \times \text{Uncertainty} + \beta \times \text{DriftScore} + \gamma \times \text{ExpectedCost}$$

Unlike traditional approaches where these weights are manually defined, the proposed framework learns the parameters α , β , and γ automatically using a meta-learning decision model trained on validation data.

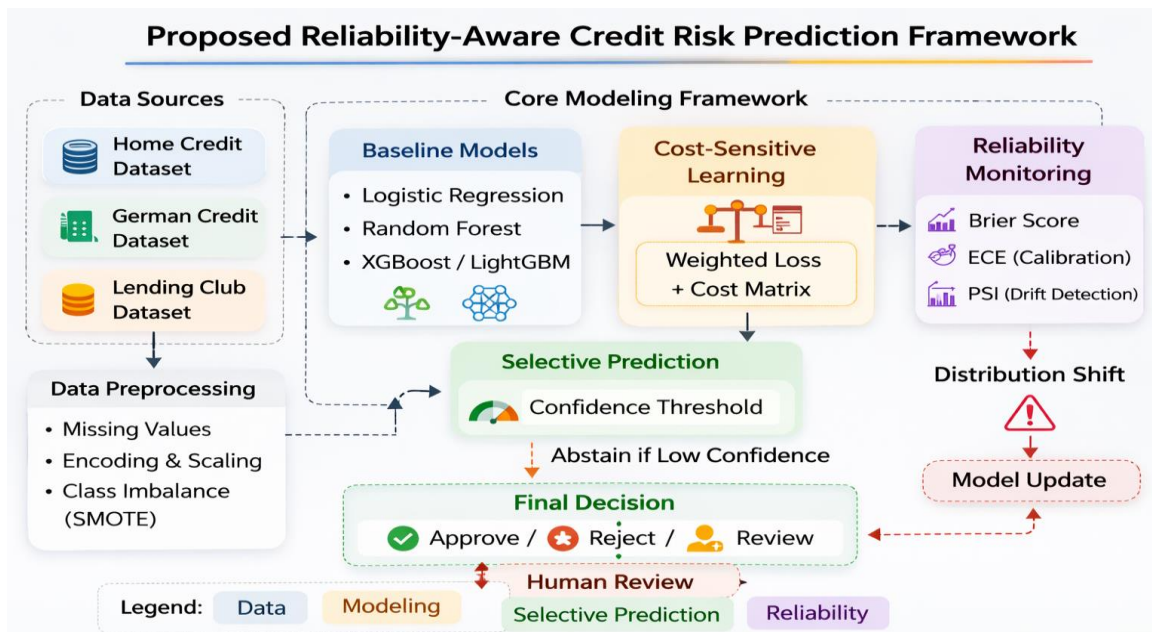
The objective of this meta-decision model is to minimize overall financial risk while maintaining reliable predictions.

Based on the calculated Review Score, the system follows a decision policy:

- **If the ReviewScore is below a predefined threshold**, the model issues an automated decision (loan approval or rejection).
- **If the ReviewScore exceeds the threshold**, the case is deferred to human experts for further evaluation.

This adaptive mechanism allows the model to abstain from unreliable predictions and ensures that high-risk decisions receive additional human verification.

By dynamically combining uncertainty estimation, distribution shift detection, and financial risk assessment, the proposed framework improves decision reliability and reduces potential financial losses in real-world credit risk prediction systems.



4 Expected Contribution

This research presents an innovative reliability-aware credit risk prediction framework that simultaneously integrates cost-sensitive learning, selective prediction, and distribution shift analysis. Unlike previous studies that have examined these components separately, this work

investigates their combined effect on model reliability, financial loss reduction, and calibration under real-world financial market conditions.

The study is expected to contribute to the field of reliable machine learning in financial systems by:

Developing a reliability-aware credit risk prediction framework that enhances trustworthiness in financial decision-making.

Integrating cost-sensitive learning and selective prediction into a unified framework to account for both economic consequences and prediction uncertainty.

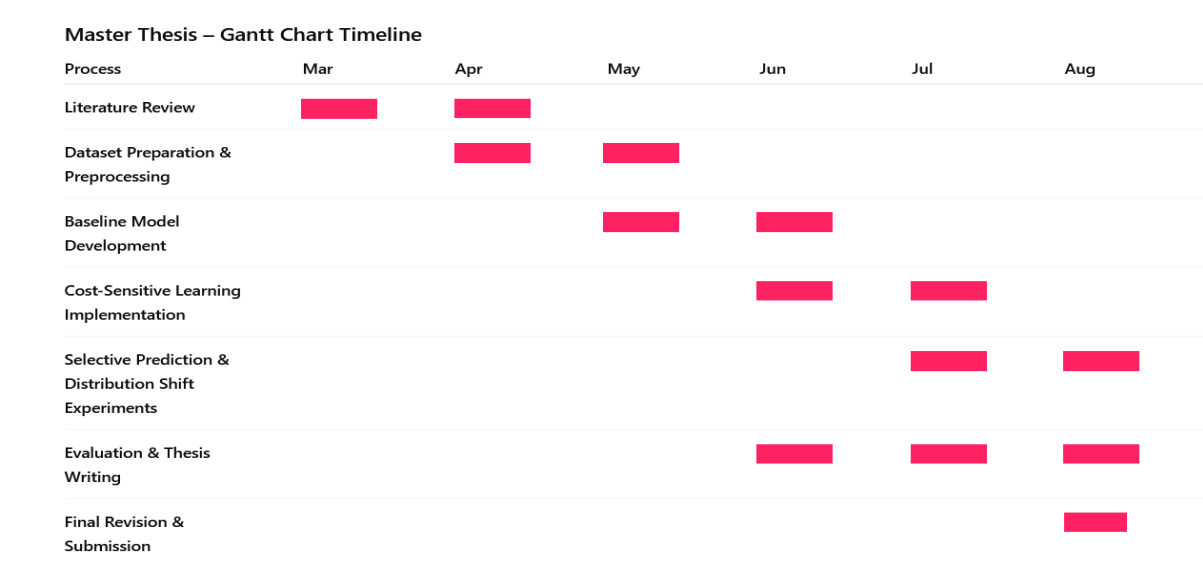
Evaluating model robustness under distribution shifts, ensuring stable performance when input data distributions change.

Introducing an adaptive decision mechanism that automatically combines uncertainty, distribution shift indicators, and financial risk signals to determine when the model should abstain from prediction and defer decisions to human experts.

Providing practical insights for deploying reliable AI systems in financial institutions, bridging research and real-world application.

5 Timeline

- Month 1 – Literature Review
- Month 2 – Dataset Preparation
- Month 3 – Baseline Model Development
- Month 4 – Cost-Sensitive Learning Implementation
- Month 5 – Selective Prediction and Distribution Shift Experiments
- Month 6 – Evaluation and Thesis Writing



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